An extremal even unimodular lattice in dimension 112: supplement

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This supplement records search motivation, additional checks, running times and the support-module argument, and classifies the even unimodular lattices of dimension 112 on which $\mathrm{SL}_2(29)$ acts as on L_{112} (§S6). Section, proposition, appendix and bibliography identifiers without an S prefix, and computation labels such as [C1], refer to the main paper. Theorem A is proved in §§2–6 of that paper. Route T (§S4.1, Theorem S4.1) proves $\min L_{112} \geq 10$ again with a different reduction (Coulangeon's inequality and an enumeration of support modules in place of Lemma 4.2 and Corollary 4.3). Its last step uses the same computations F5a and F5b as the main proof, together with a further finite check [C32] that the main proof does not use; so it checks the reduction independently, but not the computations F0–F5.

S1. Siegel means and search motivation

Write $\Sigma(n)$ for the Siegel weighted mean number of nonzero vectors of norm below the extremal bound $\mu(n)$. It is the sum of the corresponding Eisenstein-series coefficients below.

As a measure of how hard it is to find an extremal lattice, we use the average over all even unimodular lattices. By Siegel's theorem this average is an explicit modular form (an Eisenstein series), so its coefficients can be computed exactly.

Up to isometry there are finitely many even unimodular lattices of dimension n (they form a single genus). Weighting each lattice Λ by $1/|\mathrm{Aut}\ \Lambda|$, Siegel's theorem (a special case of the Siegel–Weil formula) gives

$$\frac{\sum_\Lambda \theta_\Lambda/|\mathrm{Aut}\ \Lambda|}{\sum_\Lambda 1/|\mathrm{Aut}\ \Lambda|} = E_{n/2}(q) = 1 - \frac{2k}{B_k}\sum_{m\geq 1}\sigma_{k-1}(m)\,q^m \qquad (k = n/2)$$

[Kin03, Thm. 18]. Here B_k is the Bernoulli number (negative for $k \equiv 0 \bmod 4$) and $\sigma_{k-1}(m)$ the sum of the $(k-1)$-th powers of the divisors of m. The coefficient of q^m is the average number of vectors of norm $2m$. The weight $1/|\mathrm{Aut}\ \Lambda|$ can be read as the likelihood of obtaining Λ from a "random" construction (lattices with more symmetry are less likely).

The difference between the extremal theta series and $E_{n/2}$ is a cusp form. Table S1 compares the coefficients in dimensions 56 and 112.

Table S1. Numbers of vectors by norm: Siegel mean (coefficients of $E_{n/2}$) versus the extremal theta series.

n	norm	typical lattice (Siegel mean)	extremal lattice	comment
56	2	2×10^{-6}	0	inner shell
	4	275.3	0	inner shell ($\Sigma(56) \approx 275$)
	6	15 643 264.9	15 590 400	kissing number 0.3 % below the mean
	8	36 955 007 141.0	36 957 286 800	
	10	15 284 232 608 011.0	15 284 192 071 680	
112	2	4×10^{-29}	0	inner shell
	4	1.4×10^{-12}	0	inner shell
	6	0.0069	0	inner shell
	8	51 014.2	0	inner shell; the main contribution
	10	10 907 925 307.1	10 888 335 360	kissing number 0.18 % below the mean
	12	247 007 811 617 624	247 006 775 232 000	

The mean number of nonzero vectors of norm at most 6 in dimension 112 is about 0.007; the mean number of norm-8 vectors is about 51 000. Thus most of the mean count below the extremal bound comes from norm 8. These are weighted averages over lattices. §S1.1 explains how they were used to motivate the search without interpreting them as the distribution of lattices with a prescribed symmetry group.

By the formula above, with $n = 2k$ the average number of vectors of norm $2m$ is $-\frac{2k}{B_k}\sigma_{k-1}(m)$; $\Sigma(n)$ is its sum over $1 \le m \le \lfloor n/24 \rfloor$.

Table S2. Siegel means $\Sigma(n)$ of the number of inner-shell vectors (the Siegel sawtooth). Each row has a fixed extremal bound.

	$n \equiv 0 \pmod{24}$	$n \equiv 8$	$n \equiv 16$
$\mu = 4$	24: 94.8	32: 4.51	40: 0.076
$\mu = 6$	48: 4 651	56: 275	64: 9.09
$\mu = 8$	72: 262 716	80: 16 801	88: 716
$\mu = 10$	96: 15 732 086	104: 1 046 569	**112: 51 014**
$\mu = 12$	120: 9.73×10^8	128: 6.63×10^7	136: 3.51×10^6

Within a row, $\Sigma(n)$ drops by a factor of 15–20 for every 8 dimensions.

Table S3. Rounded Siegel means of the number of nonzero vectors below the extremal bound. The last column gives a rounded group-order scale for the illustrative benchmark $\frac{\Sigma(n)}{|}G| \le 0.05$; this is not a criterion for extremality.

n	$\mu(n)$	inner shells	$\Sigma(n)$	$\lvert G\rvert$ for $\lambda \le 0.05$
72	8	norms 2–6	262 716	5×10^6
88	8	norms 2–6	716	1×10^4
96	10	norms 2–8	15 732 086	3×10^8
104	10	norms 2–8	1 046 569	2×10^7
112	**10**	norms 2–8	**51 014**	$\mathbf{1 \times 10^6}$
120	12	norms 2–10	972 746 306	2×10^{10}

S1.1. Symmetry and the mean vector count

Let $\Sigma(n)$ be the Siegel weighted mean defined in §S1. For a chosen finite symmetry group G, consider the ratio $\lambda = \frac{\Sigma(n)}{|}G|$. If the stabilizers of the relevant vectors are trivial, their orbits have size $|G|$.

This suggests using the ratio to compare the scale of the mean vector count with an orbit size. The weighted mean over all even unimodular lattices does not, by itself, give the conditional mean over lattices carrying a specified group.

As a further modeling assumption, a Poisson number of orbits with mean λ would give the probability $e^{-\lambda}$ for having no such orbit. With only the scalar symmetries ± 1, the corresponding scale in dimension 112 is $e^{-\frac{\Sigma(112)}{2}} \approx e^{-25\,507}$. This model helped motivate the imposition of symmetry during the search; it is not a deduction about the actual distribution of lattices, and it estimates neither the correctness nor the running time of the proof.

Table S4. The ratio $\lambda = \frac{\Sigma(n)}{|G|}$ for selected lattices. $|G|$ is the order of the automorphism group (for gluings A, B: of the visible subgroup; for L_{112}: Theorem B, §8).

lattice	$\Sigma(n)$	$\lvert G\rvert$	λ	how it was found
P_{48n}	4 651	524 160	0.009	from a group
P_{48p}	4 651	145 728	0.03	from a code
P_{48q}	4 651	103 776	0.04	from a code
P_{48m}	4 651	1 200	3.9	exhaustive classification
Γ_{72}	262 716	5 241 600	0.05	test of all 9 structures
L_{80} (Bachoc–Nebe)	16 801	4.5×10^9	4×10^{-6}	from a group
gluings A, B (min 8)	51 014	292 320	0.17	§3
gluing C = L_{112}	51 014	1 169 280	**0.044**	this paper
(L_{112}, counting $\mathrm{SL}_2(29)$ only)	51 014	24 360	2.1	

Group orders are from [NS] and [Neb13, Thms. 3.6, 6.2]; for P_{48p} the order in [Neb13, Thm. 3.6] lacks a factor 2, as corrected on Nebe's list of publications. The last two rows illustrate that the numerical ratio changes substantially depending on whether one counts only the imposed group or the full automorphism group determined later. The proof uses the imposed group and does not require a small value of this ratio.

S2. Expected collisions in a random gluing

If f were a random parity-preserving map, the class xf of $x \in L_{56}(a)$ would be uniformly distributed among classes of the same parity. By linearity of expectation, the expected number of inner-shell vectors of L_{112} coming from type (a, b) is

$$\mathbb{E}[\#(a,b)] \approx \frac{|L_{56}(a)|\,|L_{56}(b)|}{N_{\pm}}, \qquad N_- = 2^{55} - 2^{27} \text{ (odd classes)}, \quad N_+ = 2^{55} + 2^{27} \text{ (even classes)}$$

(computation in §S2.1). This estimate is a heuristic and is not used in the proof.

Table S5. Expected numbers for a random gluing (from the theta series of L_{56}) and the actual values for gluings A, B, C.

type	expected	share	gluings A, B	gluing C
$(6,6)$	0.007	—	0	0
$(6,10)$ and $(10,6)$	13 228	26 %	0	0
$(8,8)$	37 910	74 %	292 320	0
total	51 137		292 320	0

The total agrees with the Siegel mean $\Sigma(112) = 51\,014$ within 0.2 %. Counting both inputs and partners modulo sign gives the probability scale $\frac{6\,\frac{614}{4}}{15\,590\,\frac{400}{2}} \approx \frac{1}{4}\,714$. Exhaustive tests are therefore required.

S2.1. Calculation of the expectations

Type $(6,10)$: $|L_{56}(6)||L_{56}(10)|/N_- = 15\,590\,400 \times 15\,284\,192\,071\,680/(2^{55}-2^{27}) \approx 6\,614$; type $(10,6)$ gives the same, together 13 228. Type $(8,8)$: $|L_{56}(8)|^2/N_+ \approx 37\,910$. Type $(6,6)$: $|L_{56}(6)|^2/N_- \approx 0.0067$. All three values assume that the class xf is uniformly distributed among classes of the same parity.

S3. Reproduction and additional checks

S3.1. Running times

Table S6. Main computations: elapsed time of single-threaded runs and peak memory (the values in the docstrings of the programs, Table D2).

step	method	time	memory
F1	interval enumeration, norm ≤ 4, BKZ-40 basis (v1)	34 min (8.5×10^9 nodes)	< 1 GB
F1	integer enumeration (a′)	96 min (1.5×10^{10} nodes)	—
F1	floating-point enumerations	svp 2 × (10–30 min), qfminim 30 min	—
F2	closure + orbits	finding 9 min (p2), checking 3 min (v3)	1.5 GB; 2 GB
F3	marked sums of two minimal vectors	finding 1.2 h (p3), checking 9 min (v4)	5 GB; 2 GB
F5a	odd-coset certificates (656)	finding 6 min (p4), checking 2 min (v5)	1 GB; 2 GB
F5b	canonical forms	24 min (v6); second implementation 1.7 h (both directions)	2 GB; 4 GB
F5b	CVP, radius² 2.05	4 × 9 h	—
F5b	even-coset certificates	finding 6 × 31 h, about 190 CPU hours (p5); checking 24 CPU hours, 7.9 h on 3 threads (v7)	3 GB (v7)

The computations were run on one machine (Intel Core Ultra 9 288V, 8 cores, 15 GB of memory, Linux under WSL2) with Python 3.12.15, NumPy 2.5.4 (with OpenBLAS 0.3.34), gcc 15.3.0, SageMath 10.9, fplll 5.5.0 (fpylll 0.6.4), PARI/GP 2.17.3 and GAP 4.15.1 [Soft]. Recorded checks in a second environment used Debian 13.7, Python 3.13.5, NumPy 2.5.4 and gcc 14.2. The machine was shared with other jobs, and the times above reflect its load at the time of measurement; a time with several threads is given as elapsed time together with the number of threads, and a CPU time is the sum over the threads or jobs; each verification program records its Python, NumPy and OS versions and running time in JSON. The Lean version is 4.34.1, with mathlib pinned to revision `d13f23b723b8a846827a245b89c10fc7d3f11612`.

S3.2. Verification routes

An independent verification needs only F0 (L_{112} is even unimodular) and F1–F5; the search (failed constructions, classification of gluings, choice of τ) is not needed. Each of F0–F5 can be checked in several ways, and the combination determines which criteria and programs are used and how long it takes. We do not designate one route as official; Table S7 lists them. The inputs and criteria for each route are listed below.

Baseline. Enumerating all vectors of norm ≤ 8 of L_{112} itself (dimension 112) would need about 10^{25} enumeration nodes by the Gaussian volume estimate (5.3×10^{25} with an LLL basis, 4.1×10^{24} with a BKZ-30 basis) [C16]. At 10^7 nodes per second this would take 10^{18} seconds, which is infeasible.

Table S7. Verification routes. [m] measured; [e] estimated from measured per-operation costs.

route	content	criteria and programs used	total
A (fastest)	F1 by interval enumeration (34 min [m]); F2, F3 by orbit sizes and distinct canonical forms (3 min and 9 min [m]); F5a by odd-coset certificates (2 min [m]); F5b by canonical-form look-up (24 min [m])	the interval enumerator, the canonical-form code, Proposition 6.3	about 1.2 CPU hours [m]
B (no canonical forms in the test of F5b)	as A (where F3 uses canonical forms, §6.4), with F5b by checking the distributed even-coset certificates (7.9 h on 3 threads [m]); finding them again would take 763 992 pairs (using the frame-preserving symmetry [C14]) × 0.7 s [m]	the interval enumerator, the midpoint lemma	about 25 CPU hours to check [m]; about 150 to find again [e]
C (closest vectors)	as A, with F5b by closest-vector enumeration	pruning in the enumeration (not a proof if floating point)	about 40 CPU hours [m]
D (from scratch)	rebuild the orbit lists from the four data files only	one's own code only	A–C plus 2–3 hours [e]
S ([C15], sampling; **not a proof**)	re-check k random certificates with code independent of the producer (group rebuilt from the generators, the distributed τ, 64-bit integers only)	—	$k = 200 + 200$ in 10 min [m]; **all 400 pass**
T (support modules)	reduce a vector of norm ≤ 8 to a rank-2 support module (Coulangeon's inequality), enumerate the range of Proposition S4.2 and find in each of the 631 dangerous modules a configuration excluded by F5a or F5b (§S4.1)	Coulangeon's inequality (rank 2, proved in §S4.1); the range of the enumeration (Proposition S4.2); the witnesses checked by an independent enumeration; F5a and F5b shared with the main proof. Cross-checks: the classification by invariants (two enumerations agree) and the three searches (all 763 992 representatives for test (2) and all 3 936 for test (3) checked in integers)	witnesses: check ≈ 8 min [m]; classification 1–2 min [m]; test (2): generation ≈ 6 CPU days [m], check 5.0 h on 2 threads [m]; plus F5a, F5b

The routes are alternatives for single steps of one proof, not independent proofs: all of them use F1–F3 and the deduction of §4 (route T replaces only the deduction), and routes B and C differ from A only in the test for F5b. Route S is a limited consistency check on the supplied tables: if a fraction p of entries were wrong, a sample of $k = 200$ would detect it with probability $1-(1-p)^{200}$, which is 0.87 for $p = 10^{-2}$ but only 0.001 for $p = 6.5 \times 10^{-6}$ (the rate of collisions in gluing B). Route T uses the Hermitian structure of $L_{112} = H \otimes P$, for dependent components (Step 1 of §S4.1; §S4.2 gives the same in the coordinates of the gluing) as well as for the rank-2 case. It shares the computations F5a and F5b with the main proof; the three closest-vector searches of §S4.1 are cross-checks. The Lean formalization in §7.1 covers the common deduction from the computational statements to the main conclusion, not the support-module enumeration.

Search and verification. Table S6 gives the measured costs of producing and checking each finite data set. A lower-bound verification must check the complete orbit lists and all relevant classes; a counterexample, in contrast, is verified by checking the membership and norm of one supplied vector.

S3.3. Recomputation by separately written implementations

Implementation 1 (`code/verify/`, `code/produce/`, `code/lib/`) constructed L_{112} and performed the main computations. Implementation 2 (`code/crosscheck/`) was written by two further sessions without reading the code of implementation 1. Its part `second_impl/` re-implemented the expansion of the minimal-vector orbits with their disjointness and design identities [C8], the design identities for the norm-8 orbits [C9], the check of the odd-coset certificates [C10], the canonical-form test, and a sample of 180 pairs by exhaustive enumeration in the lattice $\langle 2L_{56}, t\rangle$ with PARI, with the 10 colliding pairs of gluing B as positive control [C11]. Its part `all_data/` re-checked even unimodularity and the group action on the 112-dimensional Gram matrix, the minimum of L_{56} by the integer enumeration of §6.1 (a′), the orbit lists and all certificates [C27]. Some parts of `code/crosscheck/` (exit codes, input hashes, a few guards, and the adaptation of some checks to the published data files) were added later, with knowledge of implementation 1.

What the implementations share.

- Both: the four data files, and authorship by sessions of the same AI system (see "Use of AI tools" in the paper) working from the same specification (Lemma 4.2, Corollary 4.3, Check 5.4). A shared misreading of the specification would therefore not show up as a disagreement. The guards against it are the positive controls (gluings A and B, Table 6), the tests in both directions τ, τ^{-1}, the frame test (§S3.4), the derivation of the reduction in Lean (§7.1), and route T (§S4), which replaces the reduction.
- Implementations 1 and 2: the verification data (orbit representatives, the centralizer `centralizer.npz`, which `second_impl/` reads instead of recomputing it, and the certificates); NumPy and its BLAS for the products that are exact by bounds on the partial sums; and, for the canonical forms, the same design (classes as 56-bit integers, byte tables, the smallest image), so that the second canonical-form test guards against coding errors but not against an error in that design. Methods that differ from implementation 1 are the integer enumeration (a′) for F1, the checker of `all_data/` (orbits named by their lexicographically smallest vector, units recomputed from `commutant.npz` with PARI) and the enumeration in $\langle 2L_{56}, t\rangle$.
- The two F5b tests of implementation 1 (canonical forms and even-coset certificates) share F1–F3 and the reduction, and differ only in the final step; route T shares F5a and F5b.

S3.4. Changes of frame

Write $L_{112} = L_{56} \otimes_{M'} \Lambda$ with $\Lambda = \{(a,b) \in M'^2 : b \equiv a\tau \bmod 2M'\}/\sqrt{2}$, a Hermitian M'-lattice of real dimension 16. A **frame** is a pair (w_1, w_2) of orthogonal elements of Λ of Hermitian norm 2. Each frame exhibits $L_{112} \supset L_{56}w_1 \perp L_{56}w_2 \cong \sqrt{2}L_{56} \perp \sqrt{2}L_{56}$ and thus L_{112} as a gluing L_φ; the class of φ in Proposition 3.3(c) determines the minimum, so it must be the same for all frames. Λ has 72 elements of norm 2 and 864 frames. For τ of class C all 864 frames give class C, and for class B all give class B [C17]. This test is inexpensive and independent of the collision computations.

S3.5. Fixed sublattices and BKZ

Elements of order 3, 5, 7 of $\mathrm{SL}_2(29)$ have fixed sublattices of L_{112} of dimension 32, 16, 16 with exact minima 12, 10, 14 (determinants $3^{32}, 5^{16}, 7^{16}$) [C18]. The fixed sublattice of an element of order 5 contains explicit vectors of norm 10, so $\min L_{112} \le 10$. In Nebe's notation [Neb13, Def. 4.1] these elements have types 3–(40, 32)–32, 5–(24, 16)–16 and 7–(16, 16)–16, which satisfy the general restrictions $s \le \min(z, d)$ and $z \equiv s \pmod 2$ of [Neb13, Lemma 4.3(a), (d)]. These fixed sublattices do not establish a lower bound for the minimum of the whole lattice: gluing B has the same fixed-sublattice minima while its own minimum is 8. Nor does BKZ reduction separate the two: applying BKZ with block sizes 20, 30, 40, 50, 60 in turn, the shortest vectors found have norms 16, 16, 16, 14, 10 for L_{112} and 16, 16, 16, 18, 10 for B. BKZ-60 thus reaches norm 10 in both lattices and misses the norm-8 vectors of B (the values depend on the starting basis; those of [C18]).

S3.6. Controls and closest-vector searches

Every exhaustive test above was also run on the positive controls with known answers (Table 6). In particular, the tests detect *rare* collisions: 10 out of 1 527 984 pairs for B. The CVP test agrees with the canonical-form test on all 46 colliding pairs of gluings A and B and on samples of 150 pairs for τ and for gluing B [C34].

CVP. The class $t + 2L_{56}$ ($t = x_i u\tau$) contains a vector of norm ≤ 8 iff some $y \in L_{56}$ satisfies $\langle y + t/2, y + t/2\rangle \le 2$, a closest-vector problem in L_{56} with target $-t/2$. We ran fplll's enumeration at squared radius 2.05 for all 1 527 984 pairs (four jobs of about 9 hours, without canonical forms or K_8): **0 collisions** [C12]. The same enumeration found the norm-8 partner in every one of the 46 colliding pairs of gluings A and B [C34]. The radius can be reduced to this value because a norm-8 element corresponds to squared distance exactly 2, and the next possible norm, 12, to 3.

S3.7. Scope of the checks

Table S8. Possible errors and the checks that would detect them.

possible error	would be detected by
Orbit lists incomplete (shared by all $(8,8)$ tests)	orbit-size sums (exact); recount from K_8; design identities of degree 2, 4, 6
The canonical-form criterion is wrong	Proposition 6.3 (proof); CVP test without canonical forms; even-coset certificates for all pairs; support-module route (§S4.1)
A gap in the logic (missing type, case $x = 0$, wrong direction)	written proofs (§§3–5); two implementations re-deriving the reduction; tests in both directions τ, τ^{-1}; the formalization in Lean of the deduction from the stated hypotheses (§7.1; it does not cover Proposition 6.3, Appendix C.3 or the completeness arguments)
The 112-dimensional Gram matrix is not the lattice described	two implementations rebuilt (3.1) independently; the distributed reduced Gram matrix is checked against (3.1) in integers, through a change of basis of determinant ± 1 [C2]; frame test
$\min L_{56} < 6$	**rigorous interval enumeration** and the integer enumeration (a′); reduction to $L_{56,2}(M)$; three floating-point enumerations; exact agreement of shell sizes and design identities (indirect)
Assignment of minima to classes inconsistent	frame-change test (864 frames)
The tests are blind to rare collisions	positive controls A, B (10 in 1.5 million detected)
An error in a shared computational hypothesis	the corresponding input, completeness and arithmetic checks in §§6–7; Lean checks the deduction from these hypotheses, not their computational verification (§7.1)
An error of the platform (BLAS, compiler, hardware)	the interval enumeration (a) and the integer enumeration (a′) of F1, compiled separately; runs in two environments (§S3.1); implementation 2 (its own code, but the same NumPy and BLAS)
Altered or corrupted verification data	not a soundness issue: every property used by the proof is checked again; the SHA-256 hashes (Appendix D) record which files were checked

S4. Support modules (route T)

S4.1. Rank-two support modules

Gluings B and C use the same base lattice and differ only in the choice of τ. For a vector $z = e_1 \otimes x + e_2 \otimes y$ of $H \otimes P$ define its **support module** $N_z = \mathcal{O}x + \mathcal{O}y \subset P$. In this section we use the trace-normalized Hermitian form $k_P = h_P/2$, for which $\langle x, x\rangle = \mathrm{Tr}_{K/\mathbb{Q}} k_P(x,x) = 2k_P(x,x)$, and $k_R = h_H \otimes k_P$; determinants $\det_k$ of support modules and Table S9 refer to k_P. Coulangeon's inequality for tensors of rank 2 gives $k_R(z,z) \geq 2\sqrt{\det h_H \cdot \det_k N_z}$ [Cou00, CN13], so a vector of norm 8 ($k_R(z,z) = 4$) needs $\det_k N_z \leq 4$. For 20 symmetry-inequivalent norm-8 vectors of B the support modules are of two kinds, both saturated and both free of minimal vectors (Table S9) [C19].

Table S9. Support modules of the norm-8 vectors of gluing B.

type	reduced Gram matrix of k_P	$\det_k$	count (of 20)
I	$\begin{pmatrix} 4 & -2\sqrt{-5} \\ 2\sqrt{-5} & 6 \end{pmatrix} = 2H^{-1}$	4 (equality)	12
II	$\begin{pmatrix} 4 & \frac{9}{5}\sqrt{-5} \\ -\frac{9}{5}\sqrt{-5} & 5 \end{pmatrix}$	19/5	8

These matrices describe the form in a K-basis of the support, rather than an integral $\mathcal{O}$-basis. In particular the rational entries for type II do not specify the integral module structure, which is retained separately.

Type I is exactly twice the inverse of H, the configuration in which Coulangeon's bound is attained. This parallels [CN13, Thm. 5.3]: there the vectors of norm 3 of the Hermitian lattices R_i correspond to sublattices of P_i isometric to the Barnes lattice P_b, so that $\min R_i = 4$ if and only if P_i contains no such sublattice.

Route T. The following argument proves $\min L_{112} \geq 10$ again. It replaces the reduction of §4 (Lemma 4.2 and Corollary 4.3) by Coulangeon's inequality and an enumeration of support modules (Proposition S4.2). Its last step uses the same finite computations F5a and F5b as the main proof, together with the finite check of Proposition S4.3 [C32], which is new. Route T is therefore an independent check of the reduction, not of the computations F0–F5.

Theorem S4.1. Assume F0 (the Gram matrix of L_{56} is even integral), F1 ($\min L_{56} = 6$), F2–F4 (including $\tau^2 = -5$ and $\langle x\tau, y\tau\rangle = 5\langle x, y\rangle$), F5a and F5b, the isometry $H \otimes_{\mathcal{O}} P \cong L_{112}$ of Proposition 3.4, and the finite check of Proposition S4.3 [C32]. Then L_{112} has no nonzero vector of norm ≤ 8.

By F2–F4, F5a gives $\min(u\tau + 2L_{56}) \geq 14$ for every norm-6 vector u, and F5b gives $\min(x\tau + 2L_{56}) \geq 12$ for every norm-8 vector x (for $g \in S$, $(xg)\tau + 2L_{56} = (x\tau + 2L_{56})g$). Let $z = e_1 \otimes x + e_2 \otimes y \in H \otimes P$ be nonzero of norm ≤ 8, i.e. $k_{R(z,z)} \leq 4$. We write $h = k_P$ in this proof.

Step 1: dependent components. Suppose x, y are dependent over K. Let $\Lambda = P \cap Kx_1$ for a vector x_1 spanning $Kx + Ky$, let x_0 be a shortest vector of Λ, and $\mathfrak{a} = \{\alpha \in K : \alpha x_0 \in P\}$, a fractional ideal containing $\mathcal{O}$ with $\Lambda = \mathfrak{a}x_0$. Since x_0 is shortest, the nonzero elements of $\mathfrak{a}$ have norm ≥ 1. As a $\mathbb{Z}$-lattice with the norm form, $\mathfrak{a}$ has determinant $5/k^2$, $k = [\mathfrak{a} : \mathcal{O}]$, and Hermite's bound in dimension 2 [SPLAG, Ch. 1] gives $1 \leq 4/3 \cdot 5/k^2$, so $k \leq 2$; since the only integral ideal of norm 2 is $\mathfrak{p}_2 = \left(2, 1 + \sqrt{-5}\right)$, $\mathfrak{a} \in \left\{\mathcal{O}, \mathfrak{p}_2^{-1}\right\}$. Write $z = v \otimes x_0$ with $v \in \mathfrak{a}H$; then $k_{R(z,z)} = h_H(v, v) \cdot h(x_0, x_0)$ and $h(x_0, x_0) = n/2$ with $n = \langle x_0, x_0\rangle \geq 6$. Completing the square, $h_H(\alpha e_1 + \beta e_2) = 2\left|\beta + \alpha\sqrt{-5}/2\right|^2 + |\alpha|^2\ /2$. For $\mathfrak{a} = \mathcal{O}$ and $v \neq 0$ this is ≥ 2: if $\alpha = 0$ it is $2|\beta|^2 \geq 2$; if $|\alpha|^2 = 1$ then $\beta + \alpha\sqrt{-5}/2 \in \pm\sqrt{-5}/2 + \mathcal{O}$ has $|\cdot|^2 \geq 5/4$, so it is $\geq 5/2 + 1/2$; if $|\alpha|^2 \geq 4$ then $|\alpha|^2/2 \geq 2$. So $k_{R(z,z)} \geq 6$, a contradiction. For $\mathfrak{a} = \mathfrak{p}_2^{-1}$ the minimum of h_H on $\mathfrak{p}_2^{-1}H$ is 1: a value < 1 needs $|\alpha|^2 < 2$ and $\left|\beta + \alpha\sqrt{-5}/2\right|^2 < 1/4$, so ($\alpha = 0$ gives $2|\beta|^2 \geq 2$) it suffices to check $\alpha \in \left\{\pm 1, \pm\left(1 \pm \sqrt{-5}\right)/2\right\}$ and finitely many $\beta \in \mathfrak{p}_2^{-1}$. Hence $n \leq 8$. Since $\left(1 + \sqrt{-5}\right)/2 \in \mathfrak{a}$, the vector $x_0(1 + \tau)/2$ lies in L_{56} and has norm $6n/4$, which must be even; hence $n = 8$ and $x_0(1 + \tau) \in 2L_{56}$. Then $x_0 \in x_0\tau + 2L_{56}$ has norm $8 < 12$, contradicting F5b.

Step 2: reduction to a support module. Suppose x, y are independent over K, so $F = \mathcal{O}x + \mathcal{O}y$ is free with basis x, y. Let A be the Gram matrix of H and $B = \left(h(x_i, x_j)\right)$. Then $k_{R(z,z)} = \operatorname{Tr}\left(A\overline{B}\right)$. For positive definite Hermitian 2×2 matrices $A = \left(\begin{smallmatrix} a & \alpha \\ \overline{\alpha} & b \end{smallmatrix}\right)$, $C = \left(\begin{smallmatrix} c & \gamma \\ \overline{\gamma} & d \end{smallmatrix}\right)$ one has

$$\operatorname{Tr}(AC) \geq 2\sqrt{\det A \det C}$$

(the rank-2 case of [Cou00, Lemma 3.3]; elementary proof: with $g = \sqrt{ab}$, $h' = \sqrt{cd}$, $m = |\alpha|$, $m' = |\gamma|$, we have $\mathrm{Tr}(AC) = ac + bd + 2\,\mathrm{Re}(\alpha\overline{\gamma}) \geq 2gh' - 2mm'$ by $ac + bd \geq 2gh'$ and $|\mathrm{Re}(\alpha\overline{\gamma})| \leq mm'$; and $gh' - mm' \geq \sqrt{(g^2 - m^2)(h'^2 - m'^2)} = \sqrt{\det A \det C}$, because $g > m$, $h' > m'$ by positive definiteness and the difference of the squares is $(gm' - mh')^2$). With $\det A = 1$ this gives $\det_k F \leq 4$. The saturation $N = P \cap (Kx + Ky)$ contains F with finite index, so $\det_k N = \det_k F/[N : F] \leq 4$ (here $[N : F]$ is the index of abelian groups, which for Hermitian lattices enters to the first power; for non-free N the determinant is taken with respect to a pseudo-basis, including the norm of the coefficient ideal). Thus N is a τ-stable rank-2 $\mathcal{O}$-submodule of P with even integral trace form, real minimum ≥ 6, $\det_k N \leq 4$, and $z \in H \otimes N$; we call such N *dangerous* if $H \otimes N$ has a nonzero vector of norm ≤ 8.

Proposition S4.2 (range of the enumeration). Every such N has a pseudo-basis $N = \mathfrak{A}x \oplus \mathfrak{B}y$ with $\mathfrak{A}, \mathfrak{B} \in \{\mathcal{O}, \mathfrak{p}_2^{-1}\}$, $a = h(x, x) \in \{3, ..., 6\}$, an integer $b = h(y, y) \geq a$, $c = h(x, y) = p/2 - (q/10)\sqrt{-5}$ with $p, q \in \mathbb{Z}$, $|p| \leq ma$ and $|q| \leq 5ma$ ($m = 2$ for $(\mathfrak{A}, \mathfrak{B}) = (\mathcal{O}, \mathfrak{p}_2^{-1})$, $m = 1$ otherwise), and $N(\mathfrak{A})N(\mathfrak{B})(ab - |c|^2) \leq 4$.

Proof. Take x a shortest vector of N. As in Step 1, $N \cap Kx = \mathfrak{A}x$ with $\mathfrak{A} \in \{\mathcal{O}, \mathfrak{p}_2^{-1}\}$. The quotient $N/\mathfrak{A}x$ is torsion-free of rank 1, hence projective, so $N = \mathfrak{A}x \oplus \mathfrak{B}'y'$ (Steinitz); rescaling y' by an element of K we may take $\mathfrak{B}$ to be one of the two class representatives $\mathcal{O}, \mathfrak{p}_2^{-1}$. Then $y \in N$, so b is an integer and $b \geq a$. The substitution $y \mapsto y + \gamma x$, $\gamma \in \mathfrak{A}\mathfrak{B}^{-1}$, keeps the pseudo-basis and changes c to $c + \overline{\gamma}a$; since $\mathfrak{A}\mathfrak{B}^{-1}$ is one of $\mathcal{O}$, $\mathfrak{p}_2$, $\mathfrak{p}_2^{-1}$ and contains $m\mathcal{O}$ ($\mathfrak{p}_2 \supset 2\mathcal{O}$), c can be moved by elements of $ma\mathcal{O}$ into the rectangle $|\mathrm{Re}\, c| \leq ma/2$, $|\mathrm{Im}\, c| \leq ma\sqrt{5}/2$. Integrality of $\langle x, y\rangle = 2\,\mathrm{Re}\, c$ and of $\langle x\tau, y\rangle$ gives the form of c, and the rectangle gives the bounds on p, q. The real minimum $2a$ and the real determinant $400(\det_k N)^2 \leq 6400$ give $(2a)^4 \leq \gamma_4^4 \cdot 6400 = 4 \cdot 6400$ by Hermite's constant $\gamma_4^4 = 4$ [SPLAG, Ch. 1], so $a \leq 6$; the minimum 6 gives $a \geq 3$. Finally $\det_k N = N(\mathfrak{A})N(\mathfrak{B})(ab - |c|^2)$. □

Proposition S4.3 (final step). Every dangerous N in the range of Proposition S4.2 contains either (s1) x, w with $\langle x, x\rangle = \langle w, w\rangle = 8$ and $w \equiv x\tau \pmod{2N}$, or (s3τ) u, w with $\langle u, u\rangle = 6$, $0 < \langle w, w\rangle \leq 10$ and $w \equiv u\tau \pmod{2N}$. [C32]

The range contains 63 996 parameter tuples (a superset: x need not be shortest and the same N may occur several times), of which 1 171 give valid modules and 631 are dangerous. Each of the 631 has a witness in `certificates/route_t_exclusion.json`, 541 of form (s1) and 90 of form (s3τ). The checker enumerates the range itself, builds the real Gram matrix and τ itself, decides minimum and dangerousness by its own short-vector enumeration, and accepts only these two forms; the witnesses were produced by a different enumeration (exact rational Fincke–Pohst) [C32].

End of the proof of Theorem S4.1. The module $N \subset P$ of Step 2 has a pseudo-basis as in Proposition S4.2, and the map from the coordinates of the witness to N is a τ-equivariant isometry that carries $2\mathbb{Z}^4$ onto $2N \subset 2L_{56}$. A witness of form (s1) gives a norm-8 vector x with $w \in x\tau + 2L_{56}$ of norm 8, contradicting F5b; a witness of form (s3τ) gives a norm-6 vector u with $w \in u\tau + 2L_{56}$ of norm ≤ 10, contradicting F5a. □

Cross-checks. The following computations are not needed for Theorem S4.1.

- *Classification by invariants.* All modules with even integral trace form, real minimum ≥ 6 and $\det_k N \leq 4$, free and non-free, were also enumerated and grouped by invariants (determinant, theta coefficients up to norm 16, minimum and numbers of short vectors of $H \otimes N$): **97 invariant types, 63 of them dangerous** ($H \otimes N$ has a vector of norm < 10; 30 with minimum 6 and 33

with minimum 8). This first enumeration starts from a vector x ($h(x) = a$) that is shortest among the vectors extending to a basis of a free submodule $F' \subset N$ of index ≤ 2. If the true shortest vector x_0 ($h(x_0) = a_0 \leq 8$) does not extend, then $F' = \mathfrak{p}_2^{-1}x_0 \oplus \mathfrak{p}_2 w$; this decomposition need not be orthogonal, and reducing the cross term $h(x_0, w)$ modulo $(a_0/2)\mathcal{O}$ and taking $x = x_0 + \beta w$, $\beta = \pm\big(1 + \sqrt{-5}\big)$, gives $a \leq 13a_0/4 + 48/a_0 \leq 32$. (The first public version used $a \leq 24$, derived by assuming $w \perp x_0$; with $a \leq 32$ the candidates grow from 2 160 to 3 390 and the types are unchanged.) A second enumeration, from the true shortest vector with pseudo-bases as in Proposition S4.2, uses a coarser invariant; regrouping both by that invariant gives 58 dangerous types in each [C20].

- *Three searches modulo 2.* Every dangerous candidate also contains a configuration detected by one of: (1) a norm-8 x and a norm-8 element of $x\tau + 2L_{56}$ (this is F5b); (2) a norm-8 x and an element of norm ≤ 8 (zero included) of $x(1 + \tau) + 2L_{56}$; (3) a norm-6 u and an element of norm ≤ 10 of $u\mu + 2L_{56}$, $\mu \in \{1, \tau, 1 + \tau\}$. Search (2) is not needed: preferring (s1) covers every dangerous module without it, and even when search (2) is preferred, 56 modules can be covered only by (s1), so F5b cannot be avoided. Search (3) with $\mu = \tau$ is F5a, and $\mu \in \{1, 1 + \tau\}$ are not needed. The searches were nevertheless carried out as checks. Test (2) enumerated 763 992 pairs in floating point without finding an element of norm at most 8 [C21]. Representatives have been generated for all 763 992 classes, with none unresolved, and all have been checked in integer arithmetic: for every representative w_0 the checker verifies membership in the specified class, norm in $\{12, 16, 20\}$ and $|\langle w_0, y\rangle| \leq 6$ for every minimal vector y of L_{56} [C21]. The orbit lists used to specify the classes and expand the minimal vectors are shared with the main proof (F2, F3); the inner-product check uses the same exact routine as the even-coset checker. For (3), the inputs are the 656 norm-6 S-orbit representatives times three centralizer representatives after reduction by the symmetry preserving τ: $656 \times 3 = 1\,968$ input vectors u. For $\mu = 1$, $\pm u$ already belongs to the coset, and only elements independent of u over K are relevant. For $\mu = \tau, 1 + \tau$, all $1\,968 \times 2 = 3\,936$ supplied representatives have been checked in integer arithmetic. They exclude norms at most 10 for $\mu = \tau$, and at most 8 for $\mu = 1 + \tau$; the latter cosets are even, so norm 10 is excluded by parity as well [C21].

For test (2), verification was performed on two disjoint batches of 716 104 and 47 888 classes; their union agrees row by row with the complete file. The measured elapsed times on the shared machine, using two threads, were 16 778 s and 1 368 s, about 5.0 hours in total; a single-thread trial took about 0.028 s per class, which gives about 6 CPU hours for all classes on an unloaded core. The representatives were generated by resumable single-threaded jobs on the shared machine at 0.4–1.7 s per class, depending on the load; the elapsed times of the jobs add up to about 5.0×10^5 s, about 6 days [C21]. The collisions of gluing B were detected as controls.

S4.2. Dependent components

Step 1 of §S4.1 treats dependent components in the coordinates of $H \otimes P$. This section gives the corresponding argument in the coordinates of the gluing.

Let $z = \frac{1}{\sqrt{2}}(x, v) \in L_{112}$ with $v = x\tau + 2y$, and let $K = \mathbb{Q}(\sqrt{-5})$ act on $L_{56} \otimes \mathbb{Q}$ by $\sqrt{-5} \mapsto \tau$. Suppose $y \in Kx$ and $\langle z, z\rangle \leq 8$. By the proof of Lemma 4.2, $x \neq 0$ and $v \neq 0$, and $(n, m) = (\langle x, x\rangle, \langle v, v\rangle) \in \{(6, 6), (6, 10), (10, 6), (8, 8)\}$. We show $(n, m) = (8, 8)$, i.e. z gives a collision of type $(8, 8)$ with $v = \lambda x$, $N(\lambda) = 1$. (The claim that such a z has norm ≥ 12 is false in general: for a gluing along an element of the centralizer, $v = xu$ gives the forced collisions of §3.2.)

Since $\langle x, x\tau\rangle = 0$ and $\langle x\tau, x\tau\rangle = 5n$, the map $\lambda \mapsto \lambda x$ identifies K with Kx and carries the norm form to $nN(\lambda)$. The lattice $\Lambda = L_{56} \cap Kx$ is a rank-1 $\mathcal{O}$-module, $\Lambda = \mathfrak{a}x$ for a fractional ideal $\mathfrak{a} \supset \mathcal{O}$;

put $k = [\mathfrak{a} : \mathcal{O}]$. As a $\mathbb{Z}$-lattice of rank 2, Λ is even with minimum ≥ 6 (F1) and determinant $5n^2/k^2$; Hermite's bound in dimension 2 gives $\det \Lambda \geq 3 \cdot 36/4 = 27$, so $k \leq 2, 3, 4$ for $n = 6, 8, 10$. The integral ideals of $\mathcal{O}$ of norm ≤ 4 are $\mathcal{O}$, $\mathfrak{p}_2 = \left(2, 1 + \sqrt{-5}\right)$, the two primes $\mathfrak{p}_3, \mathfrak{p}_3' = \left(3, 1 \pm \sqrt{-5}\right)$ and $\mathfrak{p}_2^2 = (2)$, so $\mathfrak{a} \in \left\{\mathcal{O}, \mathfrak{p}_2^{-1}, \mathfrak{p}_3^{-1}, \mathfrak{p}_3'^{-1}, \frac{1}{2}\mathcal{O}\right\}$. Write $v = \lambda x$; then $\lambda \in \sqrt{-5} + 2\mathfrak{a}$ (because $y \in \Lambda$) and $N(\lambda) = m/n \in \{1, 5/3, 3/5\}$.

- $\mathfrak{a} = \mathcal{O}$: $\lambda = 2p + (1 + 2q)\sqrt{-5}$ with $p, q \in \mathbb{Z}$, so $N(\lambda) \geq 5$. Impossible.
- $\mathfrak{a} = \frac{1}{2}\mathcal{O}$ (only $n = 10$): $1/2 \in \mathfrak{a}$ would give a vector of norm $n/4$, not an even integer. Impossible.
- $\mathfrak{a} = \mathfrak{p}_2^{-1}$: $2\mathfrak{a} = \mathfrak{p}_2$ and $\sqrt{-5} + \mathfrak{p}_2 = \mathcal{O} \setminus \mathfrak{p}_2$, whose norms are odd integers. Hence $N(\lambda) = 1$ and $m = n$.
- $\mathfrak{a} = \mathfrak{p}_3^{-1}$ or $\mathfrak{p}_3'^{-1}$ ($n \in \{8, 10\}$, since $k \leq 2$ for $n = 6$): $N(\lambda) \in N(\mathfrak{a})\mathbb{Z} = \frac{1}{3}\mathbb{Z}$ excludes $3/5$; the ratio $5/3$ would need $(n, m) = (6, 10)$, which these ideals do not allow. So $N(\lambda) = 1$ and $(n, m) = (8, 8)$.

In the case $\mathfrak{a} = \mathfrak{p}_2^{-1}$, $n = 6$ is impossible: the vector $x(1 + \tau)/2 \in \Lambda$ would have the odd norm 9. Hence $(n, m) = (8, 8)$ in all cases. These collisions are excluded by F5b, as in Step 1 of §S4.1.

S5. Constructions considered in the search

Before dimension 112. More than 190 candidate lattices in dimensions 96–112 (cyclotomic ideal lattices and their ideal-class twists, Hermitian tensor products over imaginary quadratic fields, gluings $48 + 48$ and 48×2) were tested first; every one had inner-shell vectors. In dimension 96 the ratio λ of the available symmetry groups was ≥ 300; in hindsight this is consistent with these failures, and it motivated the search for constructions with larger symmetry.

The first four gluings in 112 came from Hermitian tensor products $T \otimes L_{56,2}(M)$ over imaginary quadratic fields ($\mathbb{Q}(\sqrt{-11})$: one class; $\mathbb{Q}(\sqrt{-2})$: three classes); all have forced collisions of type $(6, 6)$. Generalizing these collisions gives the classification of Proposition 3.3.

Gluings A and B. Both have exactly 6 colliding S-orbits of type $(8, 8)$, giving 292 320 vectors of norm 8, which equals the order of the explicitly constructed automorphism subgroup. They are in fact isometric: enumerating the $\mathrm{SL}_2(29)$-equivariant maps between them as in §8 yields 12 equivariant isometries up to sign, one of which is distributed with the verification data and checked in integer arithmetic [C28].

Base lattices. $L_{56,2}(M)$ is taken from the entry `L_56b` of the Catalogue of Lattices [NS]. The second base lattice was obtained inside it: for each singular $X \in M/2M$ of rank 28, $L_X = L_{56,2}(M)X + 2L_{56,2}(M)$ with the form halved is $\mathrm{SL}_2(29)$-invariant; five are even unimodular, one with centralizer of order 24 and four with order 12. Our L_{56} is one of the latter. The catalog entry `L_56c` ($= L_{56,2}(\widetilde{M})$) lists generators of a subgroup of order $2^3 \cdot 3 \cdot 7 \cdot 29$ only and states that the automorphism group contains $\mathrm{SL}_2(29) \circ \tilde{S}_3$, which is consistent with a centralizer of order 12.

S6. Even unimodular lattices of dimension 112 on which $\mathrm{SL}_2(29)$ acts as on L_{112}

Table 2 of the main paper lists the equivariant gluings modulo 2 of two copies of one base lattice. This section determines all even unimodular lattices of dimension 112 on which $\mathrm{SL}_2(29)$ acts as on L_{112}, without assuming any gluing construction. The result is that there are exactly eight such lattices up to equivariant isometry. One of them is L_{112}, and six of the others contain a vector of norm at most 8. The minimum of the remaining one, X_1, is not determined in this paper. The group $\mathrm{SL}_2(29)$ also

acts on even unimodular lattices of dimension 112 through other rational representations (for instance trivially); those lattices are not considered here.

Setting. Throughout, $S = \mathrm{SL}_2(29)$, $K = \mathbb{Q}\big(\sqrt{29}\big)$, and $\mathcal{Q}$ is the quaternion algebra of §2.4 with canonical involution $q \mapsto \bar{q}$, reduced norm nrd and reduced trace trd. Let $L = L_{56,2}(M)$ be Nebe's lattice for the maximal order M for which $C_{\mathrm{Aut}(L)}(S)$ has order 24 (equivalently, M has 24 units of reduced norm 1 [C29]), $W = L \otimes \mathbb{Q}$, and $\tilde{L} = L_{56,2}\big(\widetilde{M}\big) \cong L_{56}$ [C23]. The order M acts on W from the right; W is a right $\mathcal{Q}$-vector space of dimension 7, and the adjoint of right multiplication by q for the form of L is right multiplication by $\bar{q}$ (the adjoint involution is positive, hence the canonical one). The ring $\mathrm{End}_S(L)$ (the ring M' of §2.4 for this L) contains M and is an order, hence equals M. Since L is unimodular, $\langle Lq, L\rangle \subset \mathbb{Z}$ holds exactly when $Lq \subset L^{\#} = L$, i.e. exactly for $q \in M$.

On $\mathcal{Q}^2$ we use the Hermitian form $h(\lambda,\mu) = \lambda_1\bar{\mu}_1 + \lambda_2\bar{\mu}_2$, and $\mathcal{Q}^2$ acts on W by $x\lambda = (x\lambda_1, x\lambda_2) \in W^2$, where W^2 carries the orthogonal sum of two copies of the form of W. Then $\langle x\lambda, y\mu\rangle = \langle x\, h(\lambda,\mu), y\rangle$ for $x, y \in W$. A *lattice* $\Lambda \subset \mathcal{Q}^2$ is a finitely generated left M-submodule spanning $\mathcal{Q}^2$; its dual is $\Lambda^h = \{\mu : h(\Lambda,\mu) \subset M\}$, Λ is *unimodular* if $\Lambda^h = \Lambda$, and $U(\Lambda)$ is the group of M-linear bijections of Λ preserving h. For a lattice Λ put $L \otimes_M \Lambda = \sum_{\lambda\in\Lambda} L\lambda \subset W^2$. In §8 and §S3.4, L_{112} is written as $L_{56} \otimes_{M'} \Lambda$ over $M' = \mathrm{End}_S(L_{56}) \cong \widetilde{M}$; since $L_{56} \cong L \otimes_M I$ with I as in the proof of (c) below, this corresponds to $X_7 = L \otimes_M \Lambda_7$ here.

Theorem S6.1. Let X be an even unimodular lattice of rank 112 on which S acts by isometries with $X \otimes \mathbb{Q} \cong W \oplus W$ as $\mathbb{Q}S$-modules (this holds for the action of S on L_{112}). Then:

(a) $X = L \otimes_M \Lambda$ for the unimodular lattice $\Lambda = \mathrm{Hom}_S(L, X) \subset \mathcal{Q}^2$, after an S-equivariant isometry of $X \otimes \mathbb{Q}$ onto W^2. The classes of such X up to S-equivariant isometry correspond bijectively to the isometry classes of unimodular lattices Λ for which $L \otimes_M \Lambda$ is even, and the centralizer of S in $\mathrm{Aut}(X)$ is isomorphic to $U(\Lambda)$.

(b) There are exactly 8 isometry classes of unimodular lattices $\Lambda \subset \mathcal{Q}^2$, with $|U(\Lambda)| = 1152, 12, 288, 24, 24, 20, 288, 48$ [C29]. For all of them $L \otimes_M \Lambda$ is even [C31]. Hence there are exactly 8 lattices X up to S-equivariant isometry, $X_0, \ldots, X_7$ (Table S10), and at most 8 up to isometry. Their Gram matrices, their bases in the coordinates of (3.1), and S-equivariant isometries of X_7 onto L_{112} and of X_3 onto gluing A are distributed with the verification data [C30, C31].

(c) $X_0 = L \perp L$, $X_2 = \tilde{L} \perp \tilde{L}$ and $X_6 = L \perp \tilde{L}$ have minimum 6; X_3 is gluing A = gluing B, of minimum 8; X_4 and X_5 contain vectors of norm 8 (distributed and checked [C31]), so they are not extremal; $X_7 \cong L_{112}$ has minimum 10 (Theorem A). The minimum of X_1 is not determined in this paper.

(d) For $j \neq 7$, X_j is not isometric to L_{112}, even by an isometry that does not respect S.

Independent verification for X_1. I. Gauvin has published a computer-assisted verification, not peer-reviewed, that X_1 has minimum 10 [Gau26v]. It extends the method of §§4–6 of the main paper to the gluing of L and $\tilde{L}$ modulo 2, and includes a second implementation written without the library of [Repo], runs on two machines, and a Lean check of the final deduction from the computed facts. That work calls X_1 L112.02: the Gram matrix of X_1 in [Repo] (`certificates/genus_class_1.txt`) is identical, entry by entry, to that of L112.02 in `roiban1344/sl-2-29-rank-112-extremal-lattice` (commit `d868927`). This paper does not determine the minimum of X_1, and nothing in it depends on this result.

Table S10. The 8 classes. n_1 is the number of $\pm$-pairs of $\lambda \in \Lambda_j$ with $h(\lambda,\lambda) = 1$. A *splitting modulo 2* of Λ_j is a sublattice $J \perp J'$ of index 2^8 such that $J/\sqrt{2}$ and $J'/\sqrt{2}$ are rank-1 unimodular lattices and J and J' are primitive ($J = \Lambda_j \cap \mathcal{Q}J$, $J' = \Lambda_j \cap \mathcal{Q}J'$); a summand has type b if it is isometric to M and type c if it is isometric to the other class I. The column lists the types of all splittings [C30] the search is complete, because a summand of type b contains a vector with $h = 2$ and one of type c a vector with $h \in \{2\pi, 2\bar{\pi}\}$, $\pi = \big(7+\sqrt{29}\big)/2$ (see the proof of (c)). Through a splitting, X_j is a gluing modulo 2 of L (b) and $\tilde{L} = L \otimes_M I$ (c) in the sense of Definition 3.1, extended to two different base lattices; the frames of §S3.4 are the splittings of type cc of X_7.

j	$\lvert U(\Lambda_j)\rvert$	n_1	splittings	X_j	$\min X_j$
0	1152	24	bb	$L \perp L$ (Table 2, row 1)	6
1	12	0	bc	gluing of L and $\tilde{L}$ modulo 2	10 [Gau26v]
2	288	0	cc	$\tilde{L} \perp \tilde{L}$ (Table 2, row 3)	6
3	24	0	bb, cc	gluing A = gluing B	8
4	24	0	bc	gluing of L and $\tilde{L}$ modulo 2	≤ 8
5	20	0	none	not a gluing modulo 2	≤ 8
6	288	12	none	$L \perp \tilde{L}$	6
7	48	0	cc	L_{112} (gluing C)	10

Proof of (a). *The form.* Every S-invariant symmetric bilinear form on W^2 is $\sum_{i,j}\langle x_i H_{ij}, y_j\rangle$ with $H = \bar{H}^T \in M_2(\mathcal{Q})$, since $\mathrm{End}_S(W) = \mathcal{Q}$ and the adjoint is the canonical involution. If the form is positive definite, H is diagonalizable to $\mathrm{diag}(a_1, a_2)$ with $a_i \in K$ totally positive, and $a_i = \mathrm{nrd}(q_i)$ for some $q_i \in \mathcal{Q}^\times$ (Hasse–Schilling–Maass norm theorem: $\mathrm{nrd}(\mathcal{Q}^\times)$ consists of the elements of $K^\times$ positive at the real places where $\mathcal{Q}$ ramifies, here all of them). Hence an S-equivariant automorphism of W^2 carries the form of $X \otimes \mathbb{Q}$ to the orthogonal sum, and we may assume $X \subset W^2$ with that form.

The group ring. Let A be the image of $\mathbb{Z}S$ in $\mathrm{End}(L)$ and $E = \mathrm{End}_M(L)$, the ring of endomorphisms of L commuting with the right action of M; $A \subset E$, and E is a $\mathbb{Z}$-lattice of rank $392 = 7^2 \cdot 8$ which is a pure sublattice of $\mathrm{End}_{\mathbb{Z}}(L)$. By the double centralizer theorem, $A \otimes \mathbb{Q} = \mathrm{End}_Q(W)$, so A and E are orders in the same algebra. We claim $A_p = E_p$ for every prime p (subscripts denote $\otimes\, \mathbb{Z}_p$). For $p \nmid |S|$, $\mathbb{Z}_pS$ is a maximal order of $\mathbb{Q}_pS$ and A_p is its image in a direct factor, hence a maximal order of $E_p \otimes \mathbb{Q}$ contained in the order E_p; so $A_p = E_p$. For $p \in \{2,3,5,7,29\}$, the $\mathbb{F}_p$-algebra generated by the generators of S acting on L/pL has dimension 392 [C29] since the kernel of $E \to \mathrm{End}(L/pL)$ is pE, this says $A + pE = E$, and Nakayama's lemma gives $A_p = E_p$.

Morita equivalence. Now X is a lattice in W^2 stable under A, hence under E_p at every p. Since M is a maximal order and $\mathcal{Q}$ is unramified at p, L_p is a free M_p-module of rank 7, so E_p is a full matrix ring over M_p^{op} and, by Morita equivalence, every E_p-stable lattice in W_p^2 is $L_p \otimes_{M_p} \Lambda_p$ for the left M_p-lattice $\Lambda_p = \mathrm{Hom}_{E_p}\big(L_p, X_p\big) \subset \mathcal{Q}_p^2$. Globally, $\Lambda = \mathrm{Hom}_S(L,X) = \mathrm{Hom}_E(L,X)$ (an A-linear map is E_p-linear at every p) is a left M-lattice in $\mathrm{Hom}_{\mathbb{Q}S}(W, X\otimes\mathbb{Q}) = \mathcal{Q}^2$ with localizations Λ_p, and $X = L \otimes_M \Lambda$ because both sides have the same localizations. For the dual, $X^\#$ is again S-stable, so $X^\# = L \otimes_M \Lambda'$ with $\Lambda' = \{\mu : \langle L\mu, L\lambda\rangle \subset \mathbb{Z} \text{ for all } \lambda \in \Lambda\}$; as $\langle x\mu, y\lambda\rangle = \langle x\, h(\mu,\lambda), y\rangle$, the fact recorded in the Setting gives $\mu \in \Lambda'$ iff $h(\mu,\Lambda) \subset M$ iff $h(\Lambda,\mu) \subset M$ (as $\bar{M} = M$), i.e. $\Lambda' = \Lambda^h$. Hence X is unimodular iff Λ is. An S-equivariant isometry $\sigma : X \to X'$ induces the M-linear bijection $\varphi \mapsto \sigma \circ \varphi$ from $\mathrm{Hom}_S(L,X)$ onto $\mathrm{Hom}_S(L,X')$, which preserves h because h is determined by the form of W^2; conversely an isometry $u : \Lambda \to \Lambda'$ gives the equivariant isometry $x\lambda \mapsto x\,u(\lambda)$ of $L \otimes_M \Lambda$ onto $L \otimes_M \Lambda'$. These constructions are mutually inverse, which gives the bijection of classes and $C_{\mathrm{Aut}(X)}(S) \cong U(\Lambda)$.

Proof of (b). *One genus.* At a finite place v of K, $\mathcal{Q}_v \cong M_2(K_v)$ and $M_v \cong M_2(\mathcal{O}_v)$ ($\mathcal{Q}$ is ramified only at the infinite places). Under Morita equivalence, Hermitian M_v-lattices in $\mathcal{Q}_v^2$ correspond to $\mathcal{O}_v$ -lattices in K_v^4 with an alternating form, and unimodular lattices to self-dual ones (cf. [GHY01, §8]). Any two self-dual alternating lattices are isometric (both have a symplectic basis), so all unimodular lattices $\Lambda \subset \mathcal{Q}^2$ form one genus. Following [GHY01, §8], call Λ *maximal integral* if $h(\Lambda,\Lambda) \subset M$ and Λ is maximal with this property; integrality of Λ corresponds to integrality of the alternating form, so at every finite place the maximal integral lattices are exactly the unimodular ones, and the genus above is the genus of maximal integral lattices. Indeed, for an alternating form, an integral lattice N with $N \neq N^{\#}$ can be enlarged by any $x \in N^{\#} \setminus N$ (because $\langle x,x\rangle = 0$), so maximal integral lattices are self-dual, and self-dual lattices are maximal because an integral $N' \supseteq N$ satisfies $N' \subseteq N'^{\#} \subseteq N^{\#} = N$.

Mass. We use [GHY01, Prop. 9.4] (for $S_D = \emptyset$ it coincides with [GHY01, Prop. 9.3]). Its setting is: a totally real field k of degree d, here K with $d = 2$; a quaternion algebra D over k ramified at all real places, with set S_D of ramified finite places, here $\mathcal{Q}$ with $S_D = \emptyset$; a maximal order A_D, here M; a totally definite quaternionic Hermitian space V of rank m over D, Hermitian with respect to the canonical involution, so that its unitary group G is an inner form of Sp_{2m}, here $(\mathcal{Q}^2, h)$ with $m = 2$; and the genus of maximal integral A_D-lattices in V, here the genus of all unimodular Λ by the previous paragraph. For $v \notin S_D$ the stabilizer of such a lattice in $G(K_v)$ is hyperspecial, and the proposition states

$$\mathrm{Mass} = 2^{-md} \prod_{r=1}^{m} \zeta_k(1-2r) \cdot \prod_{v \in S_D} \lambda_v,$$

where the product of local factors λ_v is empty here. The mass is defined in [GHY01, (2.2)] as the volume of $G(k) \setminus G(\mathbb{A})$ for the Haar measure that gives $\mathcal{K} = G(k \otimes \mathbb{R}) \times \prod_v \mathcal{K}_v$ volume 1, $\mathcal{K}_v$ being the stabilizer of the lattice at v. The double cosets $G(k) g_i \mathcal{K}$ correspond to the isometry classes Λ_i in the genus, and $G(k) \cap g_i \mathcal{K} g_i^{-1} = U(\Lambda_i)$ is the full unitary group, -1 included; so the mass is $\sum_i 1/|U(\Lambda_i)|$, and no factor for the center enters. Hence

$$\sum_{\Lambda} \frac{1}{|U(\Lambda)|} = 2^{-md} \prod_{r=1}^{m} |\zeta_K(1-2r)|,$$

the sum over the isometry classes in the genus (the values $\zeta_K(1-2r)$ are positive here). With $m = d = 2$, $\zeta_K(-1) = 1/2$ and $\zeta_K(-3) = 157/20$ (computed exactly from generalized Bernoulli numbers and compared with Siegel's formulas and with PARI [C29]), the mass is $157/640$. Two checks of the normalization do not depend on the classes found below. (i) In rank 1 the formula gives $2^{-2}\ |\zeta_K(-1)| = 1/8$. The rank-1 genus consists of the two classes M and I (the left ideal classes of M), whose unitary groups are the groups of units of reduced norm 1 of the right orders, of orders 24 and 12 [C29], and $1/24 + 1/12 = 1/8$. This agrees with Eichler's mass formula [Voi21, §26.1], which does not use [GHY01]: summed over the left ideal classes of M, with right orders $\mathcal{O}_i$, it gives $\sum_i 1/[\mathcal{O}_i^{\times} : \mathcal{O}_K^{\times}] = 2^{1-d}\ |\zeta_K(-1)|\ h_K = 1/4$, and the fundamental unit $\left(5+\sqrt{29}\right)/2$ of K has norm -1, so every totally positive unit is a square and $[\mathcal{O}_i^{\times} : \mathcal{O}_K^{\times}] = |\mathcal{O}_i^1|/2$. (ii) For $k = \mathbb{Q}\left(\sqrt{5}\right)$ and $m = 1$ the formula gives $2^{-2}\zeta_k(-1) = 1/120$, the reciprocal of the order of the binary icosahedral group, which is the group of units of reduced norm 1 of the icosian ring; the icosian ring has class number 1, as Eichler's formula shows in the same way ($2^{-1}\zeta_k(-1) = 1/60 = 2/120$).

Enumeration. Starting from $\Lambda_0 = M \oplus M$, Kneser neighbours at the prime $\mathfrak{p} = \left(\left(7+\sqrt{29}\right)/2\right)$ of norm 5 are generated by a random walk with a fixed seed; each neighbour is checked to be an M-module and unimodular. Isometry between two lattices is decided with PARI's `qfisom` and $|U(\Lambda)|$ with

`qfauto`, applied to the nine integer-valued forms $\mathrm{Tr}_{K/\mathbb{Q}}\,\mathrm{trd}(m_k h)$, m_k running over 1 and a $\mathbb{Z}$-basis of M (a $\mathbb{Z}$-linear map preserving all nine is M-linear and preserves h). After 62 neighbours, 8 classes with $\sum 1/|U(\Lambda_j)| = 157/640$ have been found [C29] by the mass formula there are no others. The labels $0, ..., 7$ are attached through the invariants $|U(\Lambda)|$ and the numbers of vectors with $\mathrm{Tr}\,\mathrm{trd}\,h(\lambda,\lambda) \le 14$ (`certificates/genus_classes.txt`), which are pairwise distinct; so the labels do not depend on the order of the walk, and the 8 classes are pairwise non-isometric without recourse to `qfisom`. The orders $|U(\Lambda_j)|$ are recounted independently as the numbers of S-equivariant automorphisms of X_j by the complete enumeration of §8 [C31]. Each $X_j = L \otimes_M \Lambda_j$ is distributed by a reduced Gram matrix and by a basis in the coordinates of (3.1); those coordinates are the ones of $(L_{56} \otimes \mathbb{Q})^2$ with the form $\mathrm{diag}(G, G)/2$, identified with W^2 by $y \mapsto e(y)/2$, where $e : L_{56} \to L$ is an S-equivariant embedding with $\langle ex, ey\rangle = 2\langle x, y\rangle$ [C6]. The check verifies, in integer arithmetic, that the Gram matrix is even of determinant 1, that the basis realizes it, and that the generators of S act on X_j [C31] it does not depend on e.

Proof of (c). Let I denote a representative of the rank-1 unimodular class other than M. The lattices $M \oplus M$, $I \oplus I$ and $M \oplus I$ are unimodular, so each is one of the Λ_j. The element 1 of $\mathcal{O}_K$ is not a sum of two totally positive elements of $\mathcal{O}_K$, so a vector (a, b) with $h = 1$ has $a = 0$ or $b = 0$; and a rank-1 unimodular lattice J containing λ with $h(\lambda,\lambda) = 1$ is $M\lambda \cong M$, so I has no such vector. Hence $M \oplus M$ has 48 vectors with $h = 1$ (24 $\pm$-pairs), $M \oplus I$ has 24 (12 pairs) and $I \oplus I$ none. By Table S10, $M \oplus M$ is Λ_0 ($n_1 = 24$; it is also the starting point of the walk) and $M \oplus I$ is Λ_6 ($n_1 = 12$). Since $U(I \oplus I)$ contains $U(I) \times U(I)$ and the exchange of the summands, $|U(I \oplus I)| \ge 2 \cdot 12^2 = 288$, and among the classes with $n_1 = 0$ only Λ_2 has $|U| \ge 288$; so $I \oplus I$ is Λ_2. By the rank-1 case of (a) (the proof applies verbatim to W), $\tilde{L} \cong L \otimes_M J$ with $J \in \{M, I\}$, and $J = I$ because the centralizers of S in $\mathrm{Aut}(\tilde{L})$ and $\mathrm{Aut}(L)$ have orders 12 and 24 [C2, C29]. Hence $X_0 = L \perp L$, $X_2 = \tilde{L} \perp \tilde{L}$ and $X_6 = L \perp \tilde{L}$, of minimum $\min L = \min \tilde{L} = 6$ ([C6] §6.1). (In Table 2, the classes of f containing an element of C are these orthogonal sums: $L_u \cong L_1$ by Proposition 3.3(b), and $(x, v)/\sqrt{2} \mapsto ((x + v)/2, (v - x)/2)$ maps $L_1 = \{(x, v) : v \equiv x \pmod 2\}/\sqrt{2}$ isometrically onto the orthogonal sum of two copies of the base lattice.) For the completeness of the search of splittings claimed in Table S10: I contains vectors with $h = \pi$ and with $h = \bar{\pi}$, because $\Lambda_2 = I \oplus I$ has vectors with $h(\lambda,\lambda) \in \{\pi, \bar{\pi}\}$ (those with $\mathrm{Tr}\,\mathrm{trd}\,h = 14$) and π is not a sum of two totally positive elements of $\mathcal{O}_K$; so a summand J of type c contains a vector with $h \in \{2\pi, 2\bar{\pi}\}$, and one of type b a vector with $h = 2$, and the search [C30] examines every λ with $h(\lambda,\lambda) \in \{2, 2\pi, 2\bar{\pi}\}$ and the summand $\Lambda_j \cap \mathcal{Q}\lambda$ it generates. An S-equivariant isometry of X_3 onto gluing A is distributed [C31], gluing B $\cong$ gluing A [C28], and $\min = 8$ by Table 2 [C4]. Vectors of norm 8 in X_4 and X_5 are distributed and checked in integer arithmetic [C30, C31] they were found in the fixed sublattices of elements su with $s \in S$ and $u \in U(\Lambda_j)$. The minimum of X_1 is not determined in this paper.

Proof of (d). Let $\sigma : X_j \to L_{112}$ be an isometry and $S' = \sigma S \sigma^{-1} \subset \mathrm{Aut}(L_{112})$. By Theorem B(c), S (acting diagonally) is normal in $\mathrm{Aut}(L_{112})$ with quotient of order 48; the image of $S' \cong \mathrm{SL}_2(29)$ in this quotient is a quotient of $\mathrm{SL}_2(29)$ of order at most 48, hence trivial (the only nontrivial quotients of $\mathrm{SL}_2(29)$ are $\mathrm{SL}_2(29)$ and $\mathrm{PSL}_2(29)$), so $S' \subseteq S$ and $S' = S$. Then σ conjugates $C_{\mathrm{Aut}(X_j)}(S) \cong U(\Lambda_j)$ onto $C_{\mathrm{Aut}(L_{112})}(S)$, which has order 48 by Theorem B(a). Table S10 shows $|U(\Lambda_j)| \ne 48$ for $j \ne 7$. For $j \in \{0, 2, 3, 4, 5, 6\}$, (d) also follows from (c) and Theorem A alone, since X_j contains a nonzero vector of norm at most 8; Theorem B(c), and with it the classification of finite simple groups, is needed only for $j = 1$. □

Remarks. (1) Among the five classes whose Λ_j is not an orthogonal sum of rank-1 lattices (classes 1, 3, 4, 5, 7), four have a splitting modulo 2: the two of Table 2 with minimum ≥ 8 (classes 3 and

7) and two gluings of L with $\tilde{L}$ (classes 1 and 4), which Table 2 does not contain because it glues two copies of one base lattice. Class 5 has no splitting modulo 2; it has an automorphism of order 5 commuting with S ($|U(\Lambda_5)| = 20$). (2) The proof of (b) relies on PARI for the isometry tests and the automorphism groups and on the mass formula of [GHY01], whose hypotheses are matched in the proof of (b) and whose normalization is confirmed in rank 1 against Eichler's mass formula. The evenness and unimodularity of the X_j, the action of S, the vectors of norm 8, the two isometries (for the one onto L_{112} also its commuting with S) and the identity $\sum_j 1/|U(\Lambda_j)| = 157/640$ are re-checked in integer arithmetic with numpy only [C31] the orders $|U(\Lambda_j)|$ themselves come from PARI's `qfauto` and are recounted by the complete enumeration of §8 [C31]. The deduction of (a) and (d) from these facts is not formalized in Lean.